OIL INDIA LIMITED - DAILY DRILLING REPORT (DDR #19)
FIELD OPERATIONS HEADQUARTERS, DULIAJAN, ASSAM
SYNTHETIC OPERATIONAL AUDIT DOCUMENTATION (SIH26121 PROTOTYPE)

WELL IDENTIFICATION:
Well Name: Nahorkatiya-415 | UWI: OIL-NHK-415 | Rig: OIL Rig-08 (Assam Shelf)
Spud Date: 12-Nov-2023 | 24-Hr Reported Depth: 2,365 m MD (2,342 m TVD)
Active Stratigraphic Formation: Tipam Sandstone | Hole Size: 12-1/4 inch
Mud System: Potassium Chloride / Polymer WBM | Current Mud Weight: 10.6 ppg

24-HOUR OPERATIONAL LOG & CHRONOLOGY:
00:00 - 05:30: Rotary drilling 12-1/4" section from 2,310 m to 2,340 m MD. Parameters steady:
ROP: 14.5 m/hr, WOB: 22 klbs, Surface Torque: 10.8 kft-lbs, SPP: 2,450 psi.
05:30 - 06:15: Drilled into highly porous fractured Tipam Sandstone interval at 2,340 m MD.
06:15 - 07:00: CRITICAL EVENT: Complete loss of returns observed in mud pits (Pit Level Drop: 45 bbl in 12 min).
Standpipe pressure dropped sharply from 2,450 psi to 1,680 psi.
Immediately stopped rotary table, picked up off bottom to 2,330 m MD, spaced out drillstring.
07:00 - 10:30: Mixed and pumped 50 bbl high-viscosity Loss Circulation Material (LCM) pill
containing 25 ppb medium walnut shells, 15 ppb coarse mica flakes, and 10 ppb cellulosic fiber.
10:30 - 13:00: Squeezed LCM pill into fractured sand with hesitation pumping (annular pressure 280 psi).
Observed gradual pressure stabilization. Fluid level in annulus held static.
13:00 - 17:00: Controlled circulation resumed at reduced pump rate 480 GPM (previously 650 GPM).
Reduced active mud weight from 10.6 ppg to 10.2 ppg to stay within fracture gradient window.
17:00 - 24:00: Resumed washing and reaming 2,335 m - 2,365 m with full 100% returns restored.

DRILLING HAZARD SUMMARY:
- Event: Severe Lost Circulation / Porous Depleted Sand Influx
- Total Non-Productive Time (NPT): 10.5 Hours
- Recommended Mitigation: Pre-treat active system with 15-20 ppb fine calcium carbonate prior to crossing the upper Tipam Sandstone contact. Maintain ECD strictly below 10.8 ppg.

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MUD PROPERTIES & BOTTOM HOLE ASSEMBLY (BHA) TECHNICAL REPORT

MUD PROPERTIES AT 2,365 m MD:
- Mud Type: KCl-Polymer Water Based Mud (Inhibitive Shale & Sand System)
- Density: 10.2 ppg (Reduced from 10.6 ppg following lost circulation remediation)
- Funnel Viscosity: 48 sec/qt
- Plastic Viscosity (PV): 18 cP
- Yield Point (YP): 22 lb/100 sq ft
- Gel Strength: 6 / 14 lb/100 sq ft (10 sec / 10 min)
- API Fluid Loss: 4.8 cc / 30 min
- pH: 9.4 | Chlorides: 38,000 mg/L | KCl Content: 6.5% wt

BOTTOM HOLE ASSEMBLY (BHA #4):
1. 12-1/4" PDC Bit (5 blades, 16mm cutters)
2. 8" Mud Motor (1.5 deg bent housing, 7/8 lobe)
3. Float Sub & Filter Sub (tested to 3,500 psi)
4. MWD / LWD Telemetry Collar (Directional, Gamma Ray, Resistivity)
5. 8" Non-Magnetic Drill Collars (2 joints)
6. 8" Spiral Drill Collars (6 joints)
7. Drilling Jar (Mechanical/Hydraulic double acting, placed at 2,180 m MD)
8. 5" Heavy Weight Drill Pipe (15 joints)
9. 5" Range 2 S-135 Drill Pipe to surface

REMARKS & SUPERINTENDENT RECOMMENDATION:
Offset data from Nahorkatiya-412 confirms porous thief zones at 2,335 - 2,350 m MD.
Always spot LCM pill before penetrating deeper Barail formations. Maintain standby LCM inventory on rig.